

# Introduction

- Heat exchanger - can be defined as any device that transfers heat from one fluid to another or from or to a fluid and the environment.
- Classification according to transfer process
  - (i) Direct contact type-Gas -liquid, Liquid vapor
  - (ii) Indirect contact type - Storage type, Fluidized bed
- Classification according to surface compactness
  - (i) Non -Compact  $< 400m^2/m^3$
  - (ii) Compact  $> 400m^2/m^3$
- Classification according to construction
  - (i) Tubular - Double pipe, shell and tube, pipe coils
  - (ii) Plate type - PHE, spiral
  - (iii) Extended surface - Plate fin, Tube fin
  - (iv) Regenerative - Rotory, Fixed matrix
- Classification according to flow arrangements
  - (i) Single pass- Counter flow, parallel flow, cross flow
  - (ii) Multi pass - shell and tube m-shell pass, n-tube pass, plate

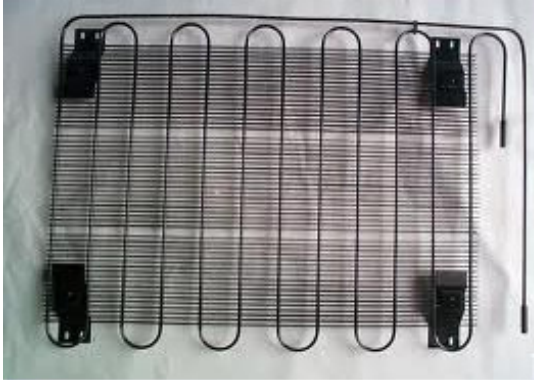
# Heat Exchangers



## Applications

Process Industry - Refinery, Petrochemicals, Dairy, Food Industries,  
Air Conditioning

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Process Industry - Refinery, Petrochemicals, Automobiles, Power plants, Air conditioning, Boilers, Furnaces